

Endogenous Grid Method for Solving the Aiyagari Model with Epstein-Zin Preferences

Jiang Hongwei

1 Model Setup

1.1 Household Problem

The economy is populated by a continuum of households facing idiosyncratic income shocks. The income state is denoted by $z \in \mathcal{Z} = \{z_1, \dots, z_{N_z}\}$, which follows a Markov process with the transition matrix:

$$\Pi_{ik} = \Pr(z_{t+1} = z_k \mid z_t = z_i).$$

In each period, given cash-on-hand m and income state z_i , the household chooses consumption c and end-of-period assets a' . Given the market interest rate r and wage w , let $R = 1 + r$. The budget constraint and asset accumulation equations are:

$$m = Ra + wz, \quad a' = m - c, \quad m' = Ra' + wz'.$$

For the Endogenous Grid Method (EGM), we define two distinct grids:

- $\mathcal{A} = \{a_1, \dots, a_{N_a}\}$: **End-of-period asset grid**. This serves as the exogenously given starting point in the EGM algorithm to reverse-engineer current consumption and cash-on-hand.
- $\mathcal{M} = \{m_1, \dots, m_{N_m}\}$: **Cash-on-hand grid**. This is the standard, fixed grid used to store the policy function $c(m, z)$ and the value function $V(m, z)$.

1.2 Firm Problem

The representative firm operates a Cobb-Douglas production technology. The demand for capital is determined by aggregate capital K and aggregate labor N . Factor prices are given by their marginal products:

$$r(K) = A\alpha \left(\frac{N}{K}\right)^{1-\alpha} - \delta, \quad w(K) = A(1-\alpha) \left(\frac{K}{N}\right)^\alpha.$$

1.3 Epstein-Zin (E-Z) Preferences

Households are endowed with Epstein-Zin recursive utility, which separates the coefficient of relative risk aversion γ from the elasticity of intertemporal substitution (EIS) ψ . We define:

$$\rho = \frac{1}{\psi}, \quad \theta = \frac{1-\gamma}{1-\rho}$$

For computational convenience, we define a transformed value function $W(m, z) = V(m, z)^{1-\rho}$. In the W -space, the Bellman equation can be written as:

$$W(m, z) = \max_c \{ (1 - \beta)c^{1-\rho} + \beta\mu(m - c, z) \}$$

where $\mu(a', z)$ is the **Certainty Equivalent** that accounts for future income uncertainty, defined as:

$$\mu(a', z_i) = \left[\sum_{k=1}^{N_z} \Pi_{ik} W(Ra' + wz_k, z_k)^\theta \right]^{1/\theta}.$$

2 Derivation of the Euler Equation under Epstein-Zin Preferences

To apply the Endogenous Grid Method, we must derive the Euler equation with respect to consumption. Taking the first-order condition (FOC) of the Bellman equation with respect to c (noting that $a' = m - c$, implying $\frac{\partial a'}{\partial c} = -1$):

$$(1 - \beta)(1 - \rho)c^{-\rho} + \beta \frac{\partial \mu(a', z)}{\partial a'} (-1) = 0$$

Rearranging yields:

$$(1 - \beta)(1 - \rho)c^{-\rho} = \beta \frac{\partial \mu(a', z_i)}{\partial a'} \quad (1)$$

Next, we compute $\frac{\partial \mu}{\partial a'}$. By the definition of $\mu(a', z_i)$:

$$\frac{\partial \mu(a', z_i)}{\partial a'} = \frac{1}{\theta} \left[\sum_k \Pi_{ik} W_k'^\theta \right]^{\frac{1}{\theta}-1} \cdot \sum_k \Pi_{ik} \theta W_k'^{\theta-1} \frac{\partial W(Ra' + wz_k, z_k)}{\partial a'}$$

Let $m'_k = Ra' + wz_k$ denote next-period cash-on-hand, and $W'_k = W(m'_k, z_k)$ denote the next-period value. Applying the chain rule, $\frac{\partial m'_k}{\partial a'} = R$. The expression simplifies to:

$$\frac{\partial \mu(a', z_i)}{\partial a'} = \mu(a', z_i)^{1-\theta} \sum_k \Pi_{ik} W_k'^{\theta-1} W_m(m'_k, z_k) R \quad (2)$$

Applying the **Envelope Theorem** to the Bellman equation with respect to the current state m (treating c as optimal, such that $\frac{\partial a'}{\partial m} = 1$):

$$W_m(m, z) = \beta \frac{\partial \mu(a', z)}{\partial a'} (1)$$

Combining this with equation (1), the marginal value of cash-on-hand W_m can be expressed in terms of current marginal utility:

$$W_m(m, z) = (1 - \beta)(1 - \rho)c^{-\rho} \quad (3)$$

Substituting the future envelope condition $W_m(m'_k, z_k) = (1 - \beta)(1 - \rho)(c'_k)^{-\rho}$ into equation (2):

$$\frac{\partial \mu}{\partial a'} = \mu(a', z_i)^{1-\theta} \sum_k \Pi_{ik} W_k'^{\theta-1} (1 - \beta)(1 - \rho)(c'_k)^{-\rho} R$$

Plugging this result back into the FOC (1):

$$(1 - \beta)(1 - \rho)c^{-\rho} = \beta \left[\mu(a', z_i)^{1-\theta} \sum_k \Pi_{ik} W_k'^{\theta-1} (1 - \beta)(1 - \rho)(c'_k)^{-\rho} R \right]$$

Canceling the constant term $(1 - \beta)(1 - \rho)$ from both sides yields the **Epstein-Zin Euler equation**:

$$c^{-\rho} = \beta R \mu(a', z_i)^{1-\theta} \left[\sum_{k=1}^{N_z} \Pi_{ik} W_k'^{\theta-1} (c'_k)^{-\rho} \right] \quad (4)$$

We define the expectation term in the brackets as $\Xi(a', z_i) = \sum_k \Pi_{ik} (W_k')^{\theta-1} (c'_k)^{-\rho}$, which corresponds to the expectation computed at each EGM iteration in the algorithm.

3 Iteration Steps of the Endogenous Grid Method (EGM)

When solving household dynamic programming problems subject to borrowing constraints, traditional approaches such as Time Iteration or Howard Policy Improvement (HPI) typically predetermine the current state m and employ nonlinear solvers (or root-finding on first-order conditions) to find the optimal end-of-period asset a' , which is computationally demanding. Conversely, the Endogenous Grid Method (EGM) adopts a **backward induction** strategy: leveraging the derived Euler equation, it exogenously fixes the next-period asset a' and analytically solves for the optimal current consumption c . Subsequently, the required current cash-on-hand $m = a' + c$ is deduced, completely avoiding expensive nonlinear root-finding operations.

Given the policy function $c^{(n)}(m, z)$ and value function $V^{(n)}(m, z)$ from the previous iteration, a single EGM update proceeds through the following specific steps:

1. **Set Target End-of-Period Asset:** Select a fixed point from the asset grid $a_j \in \mathcal{A}$ as the predetermined savings decision for the current period.
2. **Determine Future Cash-on-Hand:** Given a_j and possible future income states z_k , compute the next-period cash-on-hand: $m'_{jk} = Ra_j + wz_k$.
3. **Evaluate Future Policy and Value:** Perform linear interpolation over the fixed standard grid $\mathcal{M}$ (stored from the previous iteration) to evaluate next-period consumption and value functions: $c'_{jk} = c^{(n)}(m'_{jk}, z_k)$ and $V'_{jk} = V^{(n)}(m'_{jk}, z_k)$.
4. **Utility Space Transformation:** For computational convenience, transform the value function into the W -space: $W'_{jk} = (V'_{jk})^{1-\rho}$.
5. **Compute Certainty Equivalent:** For a given current state z_i , compute the Certainty Equivalent of the future value:

$$\mu_{ji}^{(n)} = \left[\sum_{k=1}^{N_z} \Pi_{ik} (W'_{jk})^\theta \right]^{1/\theta}.$$

6. **Evaluate Expected Marginal Utility:** Compute the expectation integral on the right-hand side of the Euler equation (4):

$$\Xi_{ji}^{(n)} = \sum_{k=1}^{N_z} \Pi_{ik} (W'_{jk})^{\theta-1} (c'_{jk})^{-\rho}.$$

7. **Analytically Solve for Current Consumption:** Using the Euler equation, obtain the closed-form solution for the current consumption level required to make a_j the optimal choice:

$$c_{ji}^{\text{endog}} = \left[\beta R \left(\mu_{ji}^{(n)} \right)^{1-\theta} \Xi_{ji}^{(n)} \right]^{-1/\rho}.$$

8. **Construct the Endogenous Grid:** Deduce the corresponding current cash-on-hand state using the budget constraint:

$$m_{ji}^{\text{endog}} = a_j + c_{ji}^{\text{endog}}.$$

In the actual code implementation, to handle the liquidity constraint $a' \geq 0$, a boundary point ($m = 0, c = 0$) is prepended to the endogenous grid arrays for each state z_i . The corresponding certainty equivalent for this point is set to the first element $\mu_{0i}^{(n)}$ (representing the certainty equivalent when $a' = 0$).

9. **Grid Resampling (Interpolation to Standard Grid):** The resulting set of points $\{(m_{ji}^{\text{endog}}, c_{ji}^{\text{endog}})\}_{j=1}^{N_a}$ forms a non-uniform endogenous grid. To facilitate subsequent iterations, for each state z_i , the consumption c^{endog} and certainty equivalent $\mu_{ji}^{(n)}$ are mapped back to the uniform standard grid $\mathcal{M}$ via linear interpolation (using m^{endog} as the independent variable). This yields the updated policy function $c^{(n+1)}(m_h, z_i)$ and evaluation function $\mu^{(n)}(m_h, z_i)$.
10. **Update Value Function:** Update the current-period value function based on the Bellman equation:

$$W^{(n+1)}(m_h, z_i) = (1 - \beta) c^{(n+1)}(m_h, z_i)^{1-\rho} + \beta \mu^{(n)}(m_h, z_i),$$

$$V^{(n+1)}(m_h, z_i) = [W^{(n+1)}(m_h, z_i)]^{1/(1-\rho)}.$$

4 Distribution and General Equilibrium

Upon convergence of the household problem, the steady-state policy function $c^*(m, z)$ is obtained. For each joint state (a_j, z_i) , we calculate the current cash-on-hand and optimal savings:

$$m_{ji} = Ra_j + wz_i, \quad a'_{ji} = m_{ji} - c^*(m_{ji}, z_i).$$

The continuous optimal savings a'_{ji} are mapped to the nearest index on the asset grid: $\sigma_{ji} = \arg \min_{\ell} |a_{\ell} - a'_{ji}|$.

Using the asset transition policy σ and the income transition matrix Π , we construct the transition matrix P_σ for the joint state (a_j, z_i) :

$$P_\sigma((a_j, z_i), (a_\ell, z_k)) = \begin{cases} \Pi_{ik}, & \ell = \sigma_{ji}, \\ 0, & \ell \neq \sigma_{ji}. \end{cases}$$

The steady-state distribution g (treated as a row vector) satisfies $g = gP_\sigma$ subject to $\sum_{j,i} g_{ji} = 1$.

By integrating over z , we obtain the marginal distribution of assets $g_a(a_j) = \sum_i g_{ji}$, which allows us to compute the aggregate capital supply:

$$K^s(r, w) = \sum_j g_a(a_j) a_j.$$

We define the outer equilibrium mapping $G(K) = K^s(r(K), w(K))$. General equilibrium requires the capital market to clear, which involves finding a K such that $F(K) \equiv K - G(K) = 0$.

5 Algorithm Pseudocode

Algorithm 1 EGM for Epstein-Zin Aiyagari Equilibrium

Input: Parameters $\beta, \gamma, \psi, A, N, \alpha, \delta$; Grids $\mathcal{A}, \mathcal{M}, \mathcal{Z}$; Transition matrix Π ; Capital search bounds $[K_L, K_H]$; Tolerance ε

Output: Equilibrium K^*, r^*, w^* , Household policies c^*, V^* , Steady-state distribution g^*

- 1: Set $\rho = 1/\psi$ and $\theta = (1 - \gamma)/(1 - \rho)$
 - 2: **while** $K_H - K_L > \varepsilon$ **do**
 - 3: Propose candidate capital $K = (K_L + K_H)/2$
 - 4: Compute $r = r(K)$, $w = w(K)$, and $R = 1 + r$
 - 5: Initialize $c^{(0)}$ and $V^{(0)}$ on $\mathcal{M} \times \mathcal{Z}$
 - 6: **repeat**
 - 7: For all a_j, z_k , compute $m'_{jk} = Ra_j + wz_k$
 - 8: Interpolate to obtain $c^{(n)}(m'_{jk}, z_k)$ and $V^{(n)}(m'_{jk}, z_k)$
 - 9: Compute $W_{jk}^{(n)} = [V^{(n)}(m'_{jk}, z_k)]^{1-\rho}$
 - 10: For all a_j, z_i , compute $\mu_{ji}^{(n)}$ and $\Xi_{ji}^{(n)}$
 - 11: Analytically solve $c_{ji}^{\text{endog}} = [\beta R(\mu_{ji}^{(n)})^{1-\theta} \Xi_{ji}^{(n)}]^{-1/\rho}$
 - 12: Construct $m_{ji}^{\text{endog}} = a_j + c_{ji}^{\text{endog}}$, and prepend liquidity constraint boundary point $(0, 0)$
 - 13: Interpolate points $(m^{\text{endog}}, c^{\text{endog}})$ and $(m^{\text{endog}}, \mu^{(n)})$ back to the fixed grid $\mathcal{M}$ to obtain $c^{(n+1)}$ and $\mu^{(n)}(m, z)$
 - 14: Update $W^{(n+1)}(m, z) = (1 - \beta)(c^{(n+1)}(m, z))^{1-\rho} + \beta\mu^{(n)}(m, z)$
 - 15: Update $V^{(n+1)}(m, z) = (W^{(n+1)}(m, z))^{1/(1-\rho)}$
 - 16: Compute error $\Delta = \max\{\|c^{(n+1)} - c^{(n)}\|_\infty, \|V^{(n+1)} - V^{(n)}\|_\infty\}$
 - 17: **until** $\Delta < \varepsilon$
 - 18: Set $c^* = c^{(n+1)}$ and $V^* = V^{(n+1)}$
 - 19: For all (a_j, z_i) , compute $m_{ji} = Ra_j + wz_i$ and interpolate to get optimal policy $c^*(m_{ji}, z_i)$
 - 20: Compute asset savings $a'_{ji} = m_{ji} - c^*(m_{ji}, z_i)$ and map to nearest asset grid index σ_{ji}
 - 21: Construct joint transition matrix P_σ from policy σ and state transition Π , solve for steady state $g = gP_\sigma$
 - 22: Compute aggregate capital supply $G(K) = \sum_j (\sum_i g_{ji})a_j$, update net demand $F(K) = K - G(K)$
 - 23: **if** $F(K) > 0$ **then**
 - 24: $K_H \leftarrow K$ ▷ Capital shortage, increase capital stock guess
 - 25: **else**
 - 26: $K_L \leftarrow K$ ▷ Capital surplus, decrease capital stock guess
 - 27: **end if**
 - 28: **end while**
 - 29: **return** $K^*, r^*, w^*, c^*, V^*, g^*$ from the final iteration
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